PEARSON EDEXCEL INTERNATIONAL A LEVEL · CHEMISTRY · WCH12

06

Energetics

Enthalpy changes, calorimetry, Hess's Law and bond enthalpies.

ΔH and standard conditions; exothermic vs endothermic; enthalpy level diagrams; the five standard enthalpy changes; calorimetry with q = mcΔT; Hess's Law cycles; mean bond enthalpies.

UNIT	SECTIONS	STUDY TIME	FLASHCARDS
2 · WCH12	12	~5 h	18

COVERS **12 SPECIFICATION POINTS** — STATEMENTS QUOTED VERBATIM FROM THE SPECIFICATION

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HOW TO USE THIS DOCUMENT

Every section opens with the specification statement, worded exactly as Edexcel words it — that is what the examiner marks against. Definitions, equations and exam alerts are boxed; worked examples walk the full method step by step. Read a section, shut the page and recall its bullets from memory, then grind the topic's flashcard deck on the site.

SPEC 6.1 - 6.11 ≈ 10 MIN

Specification Map

THE SPECIFICATION SAYS

Students will be assessed on their ability to:

Every statement the specification assesses in this topic — use it as your revision checklist.

SPEC	ASSESSMENT STATEMENT
6.1	know that the enthalpy change, ΔH, is the heat energy change measured at constant pressure and that standard conditions are 100 kPa and a specified temperature, usually 298 K
6.2	know that, by convention, exothermic reactions have a negative enthalpy change and endothermic reactions have a positive enthalpy change
6.3	be able to construct and interpret enthalpy level diagrams, showing exothermic and endothermic enthalpy changes
6.4	know the definition of standard enthalpy change of: i reaction, ΔrH ii formation, ΔfH iii combustion, ΔcH iv neutralisation, ΔneutH v atomisation, ΔatH
6.5	be able to use experimental data to calculate: i energy transferred in a reaction recalling and using the expression: energy transferred (J) = mass (g) × specific heat capacity (J g^{-1} $°C^{-1}$) × temperature change (°C) ii enthalpy change of the reaction in kJ mol^{-1}
6.6	know Hess's Law and be able to apply it to: i constructing enthalpy cycles ii calculating enthalpy changes of reaction using data provided, or data selected from a table or obtained from experiments
6.7	CORE PRACTICAL 2 — Determination of the enthalpy change of a reaction using Hess's Law.
6.8	be able to evaluate the results obtained from experiments and comment on sources of error and uncertainty and any assumptions made in the experiments
6.9	understand the terms 'bond enthalpy' and 'mean bond enthalpy', and be able to use bond enthalpies to calculate enthalpy changes, understanding the limitations of this method
6.10	be able to calculate mean bond enthalpies from enthalpy changes of reaction
6.11	understand that bond enthalpy data gives some indication about which bond will break first in a reaction, how easy or difficult it is and therefore how rapidly a reaction will take place at room temperature

Further suggested practicals:

- the enthalpy change for the decomposition of calcium carbonate using the enthalpy changes of reaction of calcium carbonate and calcium oxide with hydrochloric acid
- the enthalpy change of combustion of an alcohol
- the enthalpy change of the reaction between zinc and copper(II) sulfate solution
- the enthalpy of hydration of anhydrous copper(II) sulfate

SPEC 6.1 ≈ 20 MIN

Enthalpy Change and Standard Conditions

THE SPECIFICATION SAYS

know that the enthalpy change, ΔH, is the heat energy change measured at constant pressure and that standard conditions are 100 kPa and a specified temperature, usually 298 K

ΔH is the heat energy change at constant pressure — and the conditions that make values comparable.

DEFINITION

Energetics

The study of energy transfers between reacting chemicals and their surroundings — the energy changes in reactions reflect changes at a molecular level. The study of these energy transfers is also known as **thermodynamics**.

- In chemical systems the major transfer of energy is by **heating**.
- Every chemical reaction is accompanied by some form of energy change — sometimes obvious, in many cases unnoticed.

The First Law of Thermodynamics

- Energy can be **neither created nor destroyed** — but it can be converted from one form to another.
- The energy transferred to the surroundings is not lost to the universe: it is **dissipated**, or spread through the surroundings.
- This is the **principle of conservation of energy** — the total energy content of the universe is constant.
- Two of the most important forms of energy are **chemical energy** and **heat energy**.

Chemical energy and heat energy

- **Chemical energy** is made up of two components: **kinetic energy**, a measure of the motion of particles in a substance, and **potential energy**, a measure of how strongly the particles interact with one another.
- **Heat energy** is the portion of the potential and kinetic energy of a substance that is responsible for its temperature — it is directly proportional to the absolute temperature.

Enthalpy and enthalpy change

DEFINITION

Enthalpy, H

A measure of the **heat content** of a substance at constant pressure.

- The enthalpy of a system **cannot be measured directly** — however, the **enthalpy change at constant pressure** can.

- Enthalpy change is denoted by the symbol **ΔH** ('delta H') and is expressed in **kJ mol⁻¹**.
- **Enthalpy change** is the amount of heat energy taken in or given out during any change in a system, provided the pressure is constant.
- The enthalpy of a substance depends on its **physical state** (solid, liquid or gas), the **amount of substance**, and the **temperature and pressure**.

Standard conditions

- Enthalpy changes for reactions vary slightly depending on the conditions under which the reaction is carried out.
- Chemists use **standard conditions** so that all reactions and corresponding enthalpy changes are comparable — they are as close to normal lab conditions as possible.
- Pressure of **100 kPa** and a specified temperature, usually **298 K (25 °C)**.
- Solutions at a concentration of **1 mol dm⁻³**, and all substances in their **normal physical states** at standard temperature and pressure.

PART OF THE SYMBOL	MEANING
Δ	change in
H	enthalpy
°	standard conditions (100 kPa, usually 298 K)

EXAM ALERT

Quote 100 kPa — not 1 atm

Older sources sometimes quote 1 atm (101 kPa) as the standard pressure. The Edexcel specification values **100 kPa** exactly, and a specified temperature usually **298 K**. Use those values in definitions and exam answers.

SPEC 6.2 ≈ 15 MIN

Exothermic and Endothermic Reactions

THE SPECIFICATION SAYS

know that, by convention, exothermic reactions have a negative enthalpy change and endothermic reactions have a positive enthalpy change

The sign convention — and what it means for the energy of the system.

DEFINITION

Exothermic reactions

Reactions which **liberate heat energy to the surroundings**. By convention, ΔH is **negative**.

- The enthalpy of the reactants is **greater than** the enthalpy of the products.

- The energy content of the system has **decreased** — some energy has left the system and entered the surroundings as heat, so there is a net decrease in the potential energy of the system.
- The reactants are therefore **thermodynamically unstable** compared with the products.
- Many exothermic reactions are spontaneous. However, despite being thermodynamically unstable, some exothermic reactions might not occur due to **kinetic stability** — a high activation energy is required to start the reaction. The combustion of petrol is highly exothermic but occurs only when ignited.

DEFINITION

Endothermic reactions

Reactions which **absorb heat energy from the surroundings**. By convention, ΔH is **positive**.

- The enthalpy of the reactants is **less than** the enthalpy of the products.
- The energy content of the system has **increased** — it has gained energy from its surroundings, so there is a net increase in the potential energy of the system.
- The reactants are therefore **thermodynamically stable** compared with the products.

KEY POINT

Bond making vs bond breaking

A reaction is overall **exothermic** if the energy released for bond making is **greater** than the energy absorbed for bond breaking — and overall **endothermic** if the energy released for bond making is **less** than the energy absorbed for bond breaking.

SPEC 6.3 ≈ 25 MIN

Enthalpy Level Diagrams

THE SPECIFICATION SAYS

be able to construct and interpret enthalpy level diagrams, showing exothermic and endothermic enthalpy changes

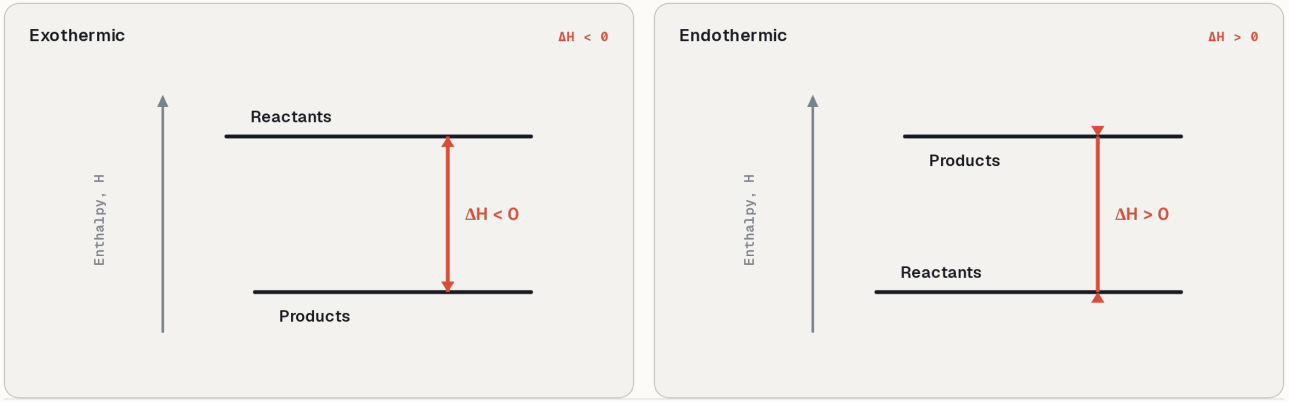
Constructing and reading the two diagrams every energetics question uses.

Two diagrams carry this whole topic: the **enthalpy level diagram** — reactant and product levels joined by the ΔH arrow — and the **reaction profile**, which additionally shows the activation energy hump. You must be able to construct and interpret both for exothermic and endothermic changes.

SPEC 6.3 · ENTHALPY LEVEL DIAGRAMS

Enthalpy level diagrams — exothermic and endothermic

construct & interpret



HOW TO READ Only the difference between the two levels matters — an enthalpy level diagram has no zero and no scale. The ΔH arrow points down for exothermic, up for endothermic.

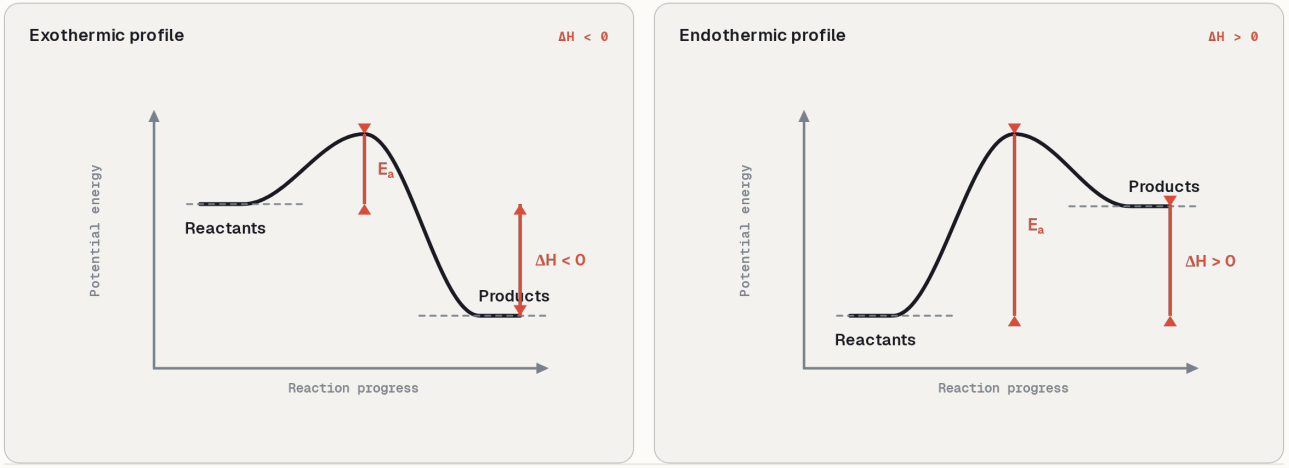
FIG — *Exothermic: products sit below the reactants, ΔH arrow points down (ΔH < 0). Endothermic: products sit above the reactants, ΔH arrow points up (ΔH > 0).*

- The vertical axis is **enthalpy, H** — there is no zero and no scale; only the **difference** between the levels matters.
- The arrow between the two levels is the enthalpy change of the reaction, with its **sign** shown by the direction.

SPEC 6.3 · REACTION PROFILES

Reaction profiles — the activation energy hump

Ea + ΔH



HOW TO READ Ea is measured from the reactant level to the top of the hump — the energy barrier that must be overcome. ΔH is measured between the reactant and product levels.

FIG — *Reaction profiles add the **activation energy, Ea** — the energy barrier that must be overcome before the reaction can proceed.*

- The **activation energy, Ea**, is measured from the reactant level up to the top of the hump.
- A **high** activation energy explains kinetic stability: an exothermic reaction with a large Ea — like petrol combustion — simply waits for a spark.

SPEC 6.4 ≈ 35 MIN

The Standard Enthalpy Changes

THE SPECIFICATION SAYS

know the definition of standard enthalpy change of:

Five definitions to know word-perfect — each with its equation.

A **thermochemical equation** is a chemical equation that includes the value of ΔH — always with **state symbols**, since a change of physical state is itself accompanied by an enthalpy change. The standard symbol ΔH° carries the ° (standard conditions); the subscripted symbols below are all quoted per mole.

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -393 \text{ kJ mol}^{-1}$$

The correct thermochemical form — burning one mole of carbon releases 393 kJ

- When a thermochemical equation is **multiplied**, ΔH is multiplied too.
- When a thermochemical equation is **reversed**, the **sign of ΔH changes** (conservation of energy).

$$C_{40}H_{82} + 60.5O_2 \rightarrow 40CO_2 + 41H_2O \qquad \Delta H = -20\,000 \text{ kJ}$$

$$2C_{40}H_{82} + 121O_2 \rightarrow 80CO_2 + 82H_2O \qquad \Delta H = -40\,000 \text{ kJ}$$

Equation ×2 → ΔH ×2

$$40CO_2 + 41H_2O \rightarrow C_{40}H_{82} + 60.5O_2 \qquad \Delta H = +20\,000 \text{ kJ}$$

Equation reversed → sign of ΔH reversed

i — Standard enthalpy change of reaction, ΔrH°

DEFINITION

ΔrH°

The enthalpy change when the number of moles of reactants **as specified in the balanced equation** react together.

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g) \qquad \Delta rH^\circ = -92 \text{ kJ mol}^{-1}$$

$$\tfrac{1}{2}N_2(g) + {}^3\!/_2 H_2(g) \rightarrow NH_3(g) \qquad \Delta rH^\circ = -46 \text{ kJ mol}^{-1}$$

The same reaction rescaled — half the equation, half the ΔH

WATCH OUT

'Per mole' means per mole of equation

ΔrH° refers to **one mole of the equation as written** — not to one mole of any particular reactant or product. Halve the equation and you halve ΔH.

ii — Standard enthalpy change of formation, ΔfH°

DEFINITION

ΔfH°

The enthalpy change that occurs when **one mole of a substance** is formed from its **elements in their standard states** under standard conditions.

$$2C(s,\ graphite) + \tfrac{1}{2}O_2(g) + 3H_2(g) \rightarrow C_2H_5OH(l)$$

Standard enthalpy of formation of ethanol

- Elements in their standard states have **zero** enthalpy of formation — ΔfH° of $H_2(g)$ = 0 and ΔfH° of $O_2(g)$ = 0.
- Carbon is usually taken as the **graphite** allotrope.

iii — Standard enthalpy change of combustion, ΔcH°

DEFINITION

ΔcH°

The enthalpy change that occurs when **one mole** of a substance is **completely burnt in excess oxygen** under standard conditions, with all reactants and products in their standard states.

- It is **always exothermic**.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$$

Standard enthalpy of combustion of ethanol

iv — Standard enthalpy change of neutralisation, ΔneutH°

DEFINITION

ΔneutH°

The enthalpy change that occurs when **one mole of water** is formed in a reaction between an acid and an alkali under standard conditions.

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l) \qquad \Delta neutH° = -57.9\ kJ\ mol^{-1}$$

- A value of about **−57 kJ mol⁻¹** is obtained when **strong** acids react with strong alkalis — strong acids and alkalis are fully ionised, so the reaction is the same in every case.

v — Standard enthalpy change of atomisation, ΔatH°

DEFINITION

ΔatH°

The enthalpy change that occurs when **one mole of gaseous atoms** is formed from the element in its standard state under standard conditions.

$$Na(s) \rightarrow Na(g) \qquad \tfrac{1}{2}Cl_2(g) \rightarrow Cl(g)$$

- It is **always endothermic**, as energy is required to break bonds.

TIP

Atomisation and bond enthalpy

For a diatomic molecule the **bond enthalpy is twice the enthalpy of atomisation**: $\tfrac{1}{2}Cl_2(g) \rightarrow Cl(g)$ is +121 kJ mol⁻¹, while $Cl_2(g) \rightarrow 2Cl(g)$ is +242 kJ mol⁻¹. The link is used in the bond-enthalpy work of 6.9.

SPEC 6.5 ≈ 40 MIN

Calorimetry: from Temperature Change to ΔH

THE SPECIFICATION SAYS

be able to use experimental data to calculate:

q = mcΔT, then scale to one mole — solution and combustion experiments.

$$\text{energy transferred (J)} = \text{mass (g)} \times \text{specific heat capacity (J g}^{-1}\ ^\circ\text{C}^{-1}) \times \text{temperature change (°C)}$$

The specification expression — recall it and use it

$$Q = m \times c \times \Delta T$$

SYMBOL	MEANING
Q	heat energy transferred, in J
m	mass of the substance to which energy is transferred, in g
c	specific heat capacity, in J g⁻¹ °C⁻¹ — for water, 4.18 J g⁻¹ °C⁻¹
ΔT	change in temperature, in °C (or K)

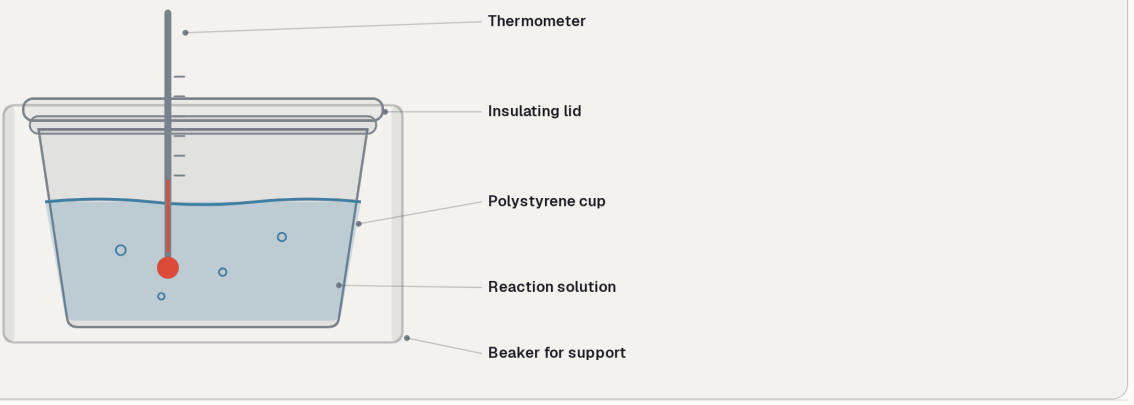
DEFINITION

Calorimeter

An insulated reaction vessel which **minimises heat exchange** with the air and the surroundings. The simplest calorimeter is a **polystyrene cup**; there are also quite complex calorimeters.

Reactions in solution (insulated container)

SPEC 6.5 · SOLUTION CALORIMETRY **The insulated-container experiment** mixing two solutions



APPARATUS A polystyrene cup inside a beaker for support — the insulated container minimises heat exchange. $q = mc\Delta T$ with m = total mass of solution, c = 4.18 J g^{-1} $°C^{-1}$, and the alkali in slight excess.

FIG — *The polystyrene cup calorimeter — the insulated container for mixing two solutions.*

- Place a measured volume of acid into the polystyrene cup and measure its temperature.
- Measure the temperature of the alkali, add it to the acid, stir with the thermometer and record the **maximum temperature reached**.
- The solution heated is the **total volume** of the two solutions — assume its density is 1 g cm^{-3} (so 1 cm^3 = 1 g) and its specific heat capacity is that of water, 4.18 J g^{-1} $°C^{-1}$.
- The alkali is often placed in slight **excess** to ensure all of the acid is neutralised.

WORKED EXAMPLE **Enthalpy change of neutralisation**

PROBLEM

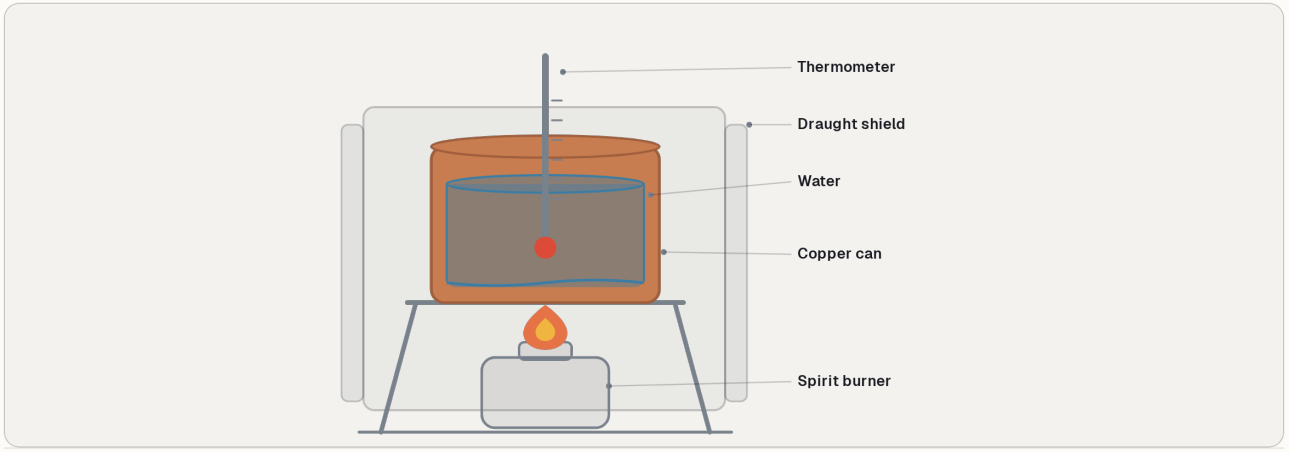
25.0 cm³ of 1.00 mol dm⁻³ acid (18.6 °C) is mixed with 25.0 cm³ of 1.20 mol dm⁻³ alkali (18.8 °C) in a polystyrene cup; the maximum temperature reached is 25.4 °C. Calculate ΔneutH°.

1. Mean starting temperature = (18.6 + 18.8) ÷ 2 = 18.7 °C, so ΔT = 25.4 − 18.7 = +6.7 K.
2. Total volume of solution heated = 25 + 25 = 50 cm³ → mass = 50.0 g (density 1 g cm⁻³).
3. Q = mcΔT = 50 × 4.18 × 6.7 = 1400.3 J = 1.4003 kJ.
4. Amount of acid neutralised = amount of water formed = (1.00 × 25.0) ÷ 1000 = 0.025 mol.
5. ΔneutH° = −1.4003 ÷ 0.025 = −56 kJ mol⁻¹.

ANSWER **ΔneutH° = −56 kJ mol⁻¹ — negative because neutralisation is exothermic**

Combustion experiments (spirit burner)

SPEC 6.5 · COMBUSTION CALORIMETRY **The spirit-burner experiment** — burning a fuel



APPARATUS The fuel burns beneath a known mass of water inside a draught shield; the copper can conducts heat to the water. Weigh the burner before and after — the mass difference is the fuel burned. Reweigh after extinguishing at about a 20 °C rise.

FIG – *Combustion calorimetry — the fuel burns beneath a known mass of water; the temperature rise of the water measures the energy released.*

- Weigh a **spirit burner** containing the fuel, place it below a beaker (or calorimeter) containing a known mass of water, and measure the water temperature.
- Light the burner and **stir the water at regular intervals**.
- When the temperature has risen by about 20 °C, put out the flame and **reweigh the burner** — the mass difference is the mass of fuel burned.

WORKED EXAMPLE **Enthalpy change of combustion of ethanol**

PROBLEM

A spirit burner burns ethanol beneath 100 cm^3 of water, raising the water from 23.0 °C to 47.6 °C. The burner falls in mass from 52.76 g to 52.37 g. Calculate $\Delta c H^\circ$ of ethanol.

1. Energy absorbed by the water: $Q = mc\Delta T = 100 \times 4.18 \times (47.6 - 23.0) = 10\,282$ J = 10.28 kJ.
2. Moles of ethanol burned: mass = 52.76 − 52.37 = 0.39 g; n = mass ÷ molar mass = 0.39 ÷ 46 = 0.00847 mol.
3. Heat produced per mole: 10.28 ÷ 0.00847 = 1213.7 kJ mol^{-1}.

ANSWER **$\Delta c H^\circ$ of ethanol = −1213.7 kJ mol^{-1} — the negative sign because combustion is exothermic**

EXAM ALERT

Sign discipline

Q comes out of the calculation **positive** because the water gained energy. The reaction **released** that energy — so the enthalpy change you report is **negative**. Forgetting the sign is the most common mark lost in 6.5.

SPEC 6.6 ≈ 40 MIN

Hess's Law and Enthalpy Cycles

THE SPECIFICATION SAYS

know Hess's Law and be able to apply it to:

The route doesn't matter — only the start and finish.

DEFINITION

Hess's Law

The enthalpy change of a chemical reaction is **independent of the path taken** in converting reactants into products, provided the initial and final conditions are the same.

- The total energy for each route is the same — diagrams showing the alternative routes are called **enthalpy cycles** (Hess's cycles).
- Hess's Law allows you to work out enthalpy changes that are **not possible to measure directly**, from other data.
- It is used for calculating: enthalpy change of reaction **from bond enthalpy**; enthalpy change of reaction **from $\Delta c H^\circ$ data**; enthalpy change of formation **from $\Delta f H^\circ$ data**.

EXAM ALERT

Why some ΔH values cannot be measured directly

The reaction may have a **high activation energy**, be **very slow**, have **more than one reaction occurring at the same time**, involve a temperature change that is **difficult to measure** — or simply **not stop** when the stated products are formed.

Route 1 — using combustion data ($\Delta c H^\circ$)

$$\Delta H[\text{reaction}] = \Sigma\ \Delta cH°[\text{reactants}] - \Sigma\ \Delta cH°[\text{products}]$$

- When the products are burned, the amount of CO_2 and H_2O produced is the same as when the reactants are burned — that is what closes the cycle.

WORKED EXAMPLE **ΔfH° of carbon monoxide — Method 1 (subtract equations)**

PROBLEM

C(s, graphite) + O_2(g) → CO_2(g) ΔcH° = −394 kJ mol^{-1}, and CO(g) + ½O_2(g) → CO_2(g) ΔcH° = −283 kJ mol^{-1}. Find ΔfH° of CO(g).

1. The formation equation is C(s, graphite) + ½O_2(g) → CO(g). It cannot be measured directly because some carbon dioxide always forms as well.
2. Reverse the second equation: CO_2(g) → CO(g) + ½O_2(g) — its ΔH becomes +283 kJ mol^{-1}.
3. Add it to the first equation; the O_2 and CO_2 cancel to give C(s, graphite) + ½O_2(g) → CO(g).
4. ΔfH°[CO(g)] = −394 + 283 = −111 kJ mol^{-1}.

ANSWER **ΔfH°[CO(g)] = −111 kJ mol^{-1}**

WORKED EXAMPLE **ΔfH° of carbon monoxide — Method 2 (Hess's cycle)**

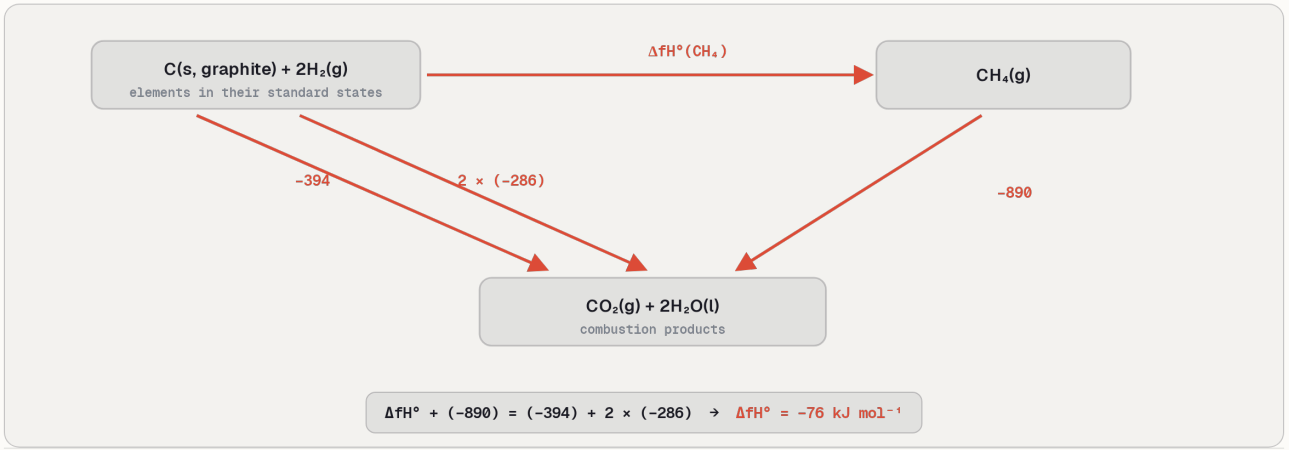
PROBLEM

The same data, laid out as an enthalpy cycle that routes both carbon species through CO_2.

1. Direct route: C(s, graphite) + ½O_2(g) → CO(g) — ΔfH°.
2. Indirect route: C(s, graphite) + O_2(g) → CO_2(g) (−394), then CO_2(g) → CO(g) + ½O_2(g) (+283).
3. Cycle equation: ΔfH° + (−283) = −394.

ANSWER **ΔfH° = −394 − (−283) = −111 kJ mol^{-1}**

SPEC 6.6 · HESS'S CYCLE – COMBUSTION DATA **Formation of methane from combustion enthalpies** route through ΔcH°



FORMULA ΔH[reaction] = Σ ΔcH°[reactants] − Σ ΔcH°[products]. When the products are burned, the amount of CO_2 and H_2O produced is the same as when the reactants are burned — that is what closes the cycle.

FIG – *ΔfH° of methane from combustion data: ΔfH° + (−890) = (−394) + 2 × (−286), so ΔfH° = −76 kJ mol⁻¹.*

WORKED EXAMPLE **ΔfH° of methane from combustion data**

PROBLEM

ΔcH° data: C(s, graphite) −394; H_2(g) −286; CH_4(g) −890 kJ mol⁻¹. Calculate ΔfH° of CH_4(g).

1. Write the formation equation: C(s, graphite) + $2H_2$(g) → CH_4(g).
2. Construct the energy cycle through the combustion products, CO_2(g) + $2H_2O$(l).
3. Apply Hess's Law to the cycle: ΔfH° + (−890) = (−394) + 2 × (−286).
4. ΔfH° = [−394 + 2(−286)] − (−890) = −966 + 890.

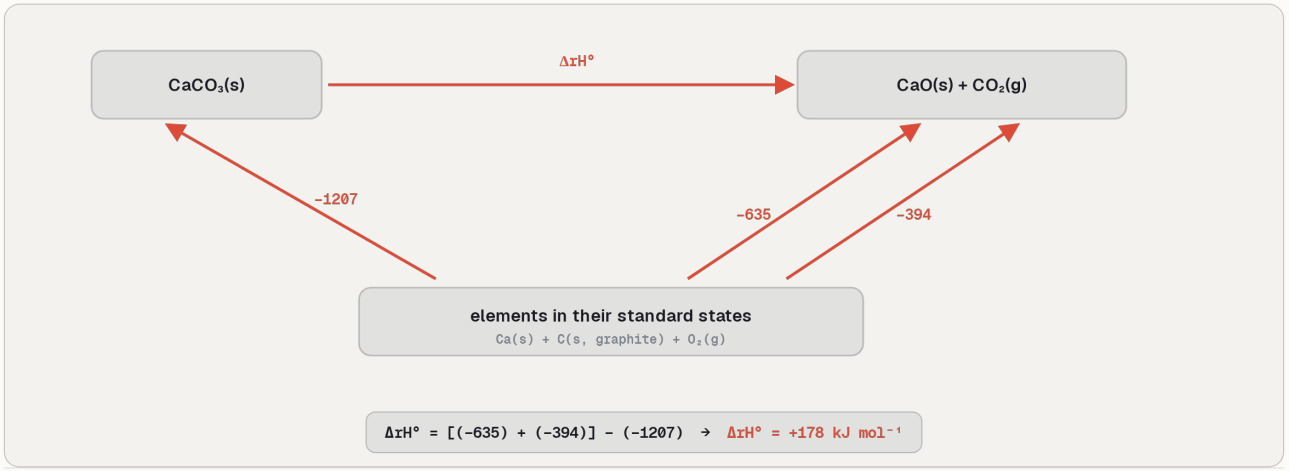
ANSWER **ΔfH°[CH_4(g)] = −76 kJ mol⁻¹**

Route 2 — using formation data (ΔfH°)

$$\Delta H[\text{reaction}] = \Sigma\ \Delta fH°[\text{products}] - \Sigma\ \Delta fH°[\text{reactants}]$$

- When enthalpies of formation for all the reactants and products are given, draw the Hess's cycle through the **elements** required to form the reactants and products.

SPEC 6.6 · HESS'S CYCLE – FORMATION DATA **Decomposition of calcium carbonate from formation enthalpies** — route through ΔfH°



FORMULA ΔH[reaction] = Σ ΔfH°[products] − Σ ΔfH°[reactants]. Draw the cycle through the elements required to form the reactants and products.

FIG – *Decomposition of calcium carbonate from formation data: ΔrH° = [(−635) + (−394)] − (−1207) = +178 kJ mol⁻¹.*

WORKED EXAMPLE **Thermal decomposition of calcium carbonate**

PROBLEM

ΔfH° data: $CaCO_3(s)$ −1207; $CaO(s)$ −635; $CO_2(g)$ −394 kJ mol⁻¹. Calculate ΔrH° for $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$.

1. Route both sides of the equation through their elements.
2. Apply Hess's Law to the cycle: ΔrH° + (−1207) = (−635) + (−394).
3. ΔrH° = [(−635) + (−394)] − (−1207) = −1029 + 1207.

ANSWER **ΔrH° = +178 kJ mol⁻¹**

SPEC 6.7 ≈ 25 MIN

Core Practical 2: ΔH via Hess's Law

THE SPECIFICATION SAYS

CORE PRACTICAL 2 — Determination of the enthalpy change of a reaction using Hess's Law.

Two measurable reactions that bracket the one you can't measure.

Core Practical 2 determines an enthalpy change of reaction that cannot be measured directly, by measuring the enthalpy changes of two reactions that **bracket** it and applying Hess's Law. Each measurable reaction is carried out in a polystyrene cup calorimeter, and q = mcΔT converts the temperature change into an enthalpy change.

WORKED EXAMPLE

Decomposition of calcium carbonate (a suggested practical)

PROBLEM

Determine Δ_rH° for $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ — a decomposition that cannot be measured directly — given: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$ $\Delta_rH^\circ = -17$ kJ mol^{-1}; $CaO(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$ $\Delta_rH^\circ = -195$ kJ mol^{-1}.

1. Both reactions share the same dissolved products, $CaCl_2(aq) + H_2O(l)$ — that is the common intermediate.
2. Reverse the second equation: $CaCl_2(aq) + H_2O(l) \rightarrow CaO(s) + 2HCl(aq)$, $\Delta H = +195$ kJ mol^{-1}.
3. Add it to the first equation; the 2HCl, $CaCl_2$ and H_2O cancel to give $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$.
4. $\Delta_rH^\circ = -17 + 195$.

ANSWER **$\Delta_rH^\circ = +178$ kJ mol^{-1}**

WORKED EXAMPLE

Hydration of anhydrous copper(II) sulfate (a suggested practical)

PROBLEM

Determine Δ_rH° for $CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4{\cdot}5H_2O(s)$, given: $CuSO_4{\cdot}5H_2O(s) + aq \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq) + 5H_2O(l)$ $\Delta_rH^\circ = +11.3$ kJ mol^{-1}; $CuSO_4(s) + aq \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq)$ $\Delta_rH^\circ = -67.0$ kJ mol^{-1}.

1. Both measured reactions dissolve the solid to the **same aqueous ions** — ‘aq’ represents the water as solvent.
2. Reverse the first equation: $Cu^{2+}(aq) + SO_4^{2-}(aq) + 5H_2O(l) \rightarrow CuSO_4{\cdot}5H_2O(s)$, $\Delta H = -11.3$ kJ mol^{-1}.
3. Add it to the second: $CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4{\cdot}5H_2O(s)$; $\Delta_rH^\circ = -67.0 - 11.3$.

ANSWER **$\Delta_rH^\circ = -78.3$ kJ mol^{-1}**

TIP

The pattern to spot

In both determinations the two measured reactions share a **common intermediate state** — the dissolved ions or the acid solution. Draw the Hess cycle through that shared state and the target enthalpy change is simply the **difference of the two measured values**.

SPEC 6.8 ≈ 30 MIN

Errors, Assumptions and Cooling Curves

> THE SPECIFICATION SAYS
>
> be able to evaluate the results obtained from experiments and comment on sources of error and uncertainty and any assumptions made in the experiments

Evaluating the experiments — and drawing the graph that corrects for heat loss.

Every calorimetry result is an evaluation opportunity: the specification expects you to comment on **sources of error and uncertainty**, state the **assumptions** made, suggest **improvements** — and to draw suitable graphs using **cooling curve corrections**.

Combustion experiments (spirit burner) — errors

- **Heat loss** to the surroundings.
- **Incomplete combustion** of the fuel — black soot is deposited on the bottom of the beaker.
- The conditions are **not standard**: water vapour is formed instead of liquid water.
- Some of the heat released in burning heats up the **air**, not the water.
- The **beaker and thermometer** absorb some of the heat produced.

Combustion experiments — assumptions

- 1 cm^3 of water is equal to **1 g**.
- **No heat is lost** to the surroundings.
- **All** the heat energy produced by the fuel is absorbed by the water.

Solution experiments (insulated container) — errors and uncertainties

- **Heat loss** to the surroundings.
- The **polystyrene cup and thermometer** absorb some of the heat produced.
- **Measurement uncertainties** involved with the use of the pipette and thermometer.

Solution experiments — assumptions

- The specific heat capacity of the resultant solution is equal to that of water — **4.18 J g^{-1} $°C^{-1}$**.
- The density of the resultant solution is equal to that of water — **1 g cm^{-3}**.
- **No heat is lost** to the surroundings.
- The total volume of the solution is equal to the **sum of the volumes** of the two solutions initially taken.

EXAM ALERT

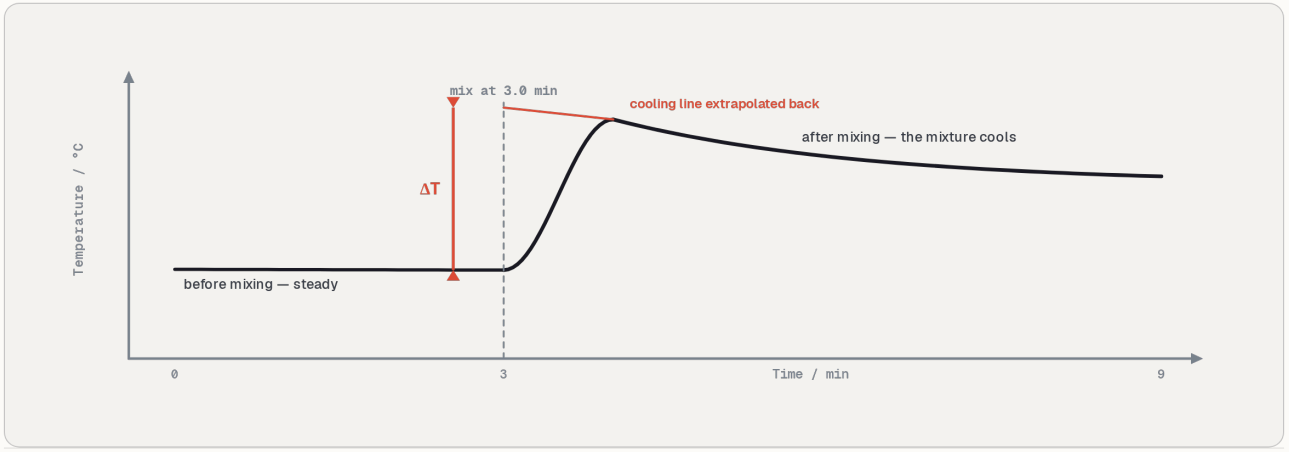
Improvements

Use a **lid** on the polystyrene cup and add **lagging** around it — both cut heat loss to the surroundings, so the measured temperature change is closer to the true value. Shielding a spirit-burner experiment from draughts works the same way.

Graphical method — cooling curves

- Take the temperature **every half minute** before mixing the reactants.
- Mix the reactants after exactly **3 minutes** (say), then keep taking readings **every half minute** as the reaction mixture cools.
- **Extrapolate** the two lines as shown and calculate the value of ΔT at the moment of mixing.
- Extrapolating the graph **compensates for heat loss** to the surroundings — this method allows for the **cooling correction**.

SPEC 6.8 · COOLING CURVE CORRECTION **The graphical method — correcting for heat loss** cooling correction



METHOD Temperature every half minute before mixing · mix at 3.0 min · keep reading as the mixture cools · extrapolate both lines back to the moment of mixing and read ΔT there — the extrapolation compensates for heat lost to the surroundings.

FIG — *The cooling correction: extrapolate the before-mixing and after-mixing lines back to the moment of mixing, and read ΔT there.*

EXAM ALERT

Why a series of readings?

Recording the temperature every half minute — rather than just a starting and a maximum value — shows the **cooling trend** clearly, which is what justifies extrapolating both lines back to the mixing time to find the corrected ΔT.

SPEC 6.9 ≈ 30 MIN

Bond Enthalpy and Mean Bond Enthalpy

THE SPECIFICATION SAYS

understand the terms ‘bond enthalpy’ and ‘mean bond enthalpy’, and be able to use bond enthalpies to calculate enthalpy changes, understanding the limitations of this method

Breaking bonds costs energy — calculating ΔH from bond enthalpies, and the limits of the method.

DEFINITION

Bond enthalpy, ΔBH

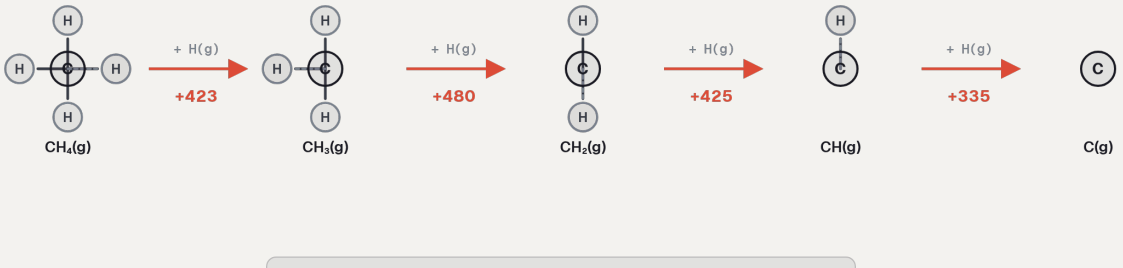
The enthalpy change when **1 mol of a bond in the gaseous state** is broken: XY(g) → X(g) + Y(g).

$$Cl_2(g) \rightarrow 2Cl(g) \qquad \Delta BH = +243 \text{ kJ mol}^{-1}$$

$$HCl(g) \rightarrow H(g) + Cl(g) \qquad \Delta BH = +432 \text{ kJ mol}^{-1}$$

- For **polyatomic** molecules, each bond must be considered separately — for example, the four C–H bonds of methane:

SPEC 6.9 · BOND ENTHALPY **The four C–H bonds of methane break one at a time** ΔBH, stepwise



E(C–H) = (423 + 480 + 425 + 335) ÷ 4 = 415.75 ≈ +416 kJ mol⁻¹

WHY MEANS Each successive bond breaks with a different enthalpy because the strength of a bond depends on its environment — so MEAN values are quoted. The C–H bond in different compounds has slightly different values, so an average is used.

FIG — *Each successive C–H bond in methane breaks with a different enthalpy — the strength of a bond depends on its environment, so **mean** values are quoted.*

DEFINITION

Mean bond enthalpy, E

The **average** quantity of energy required to break **one mole** of that covalent bond in a **gaseous species** (at 100 kPa).

$$E(C\text{–}H) \approx (423 + 480 + 425 + 335) \div 4 = +416 \text{ kJ mol}^{-1}$$

The mean C–H bond enthalpy in methane — 415.75 ≈ +416 kJ mol⁻¹

- The C–H bond in different compounds has slightly different bond enthalpy values, so an **average value** is used.
- The enthalpy change calculated from **standard enthalpy of formation or combustion data is more accurate** than one calculated from bond enthalpy values — because bond enthalpy values are means.

WATCH OUT

Limitations of the bond-enthalpy method

Bond enthalpies are measured in the **gaseous state** only, and the values are **not exact** — they are averages. Both effects make the calculated ΔH differ from the measured value.

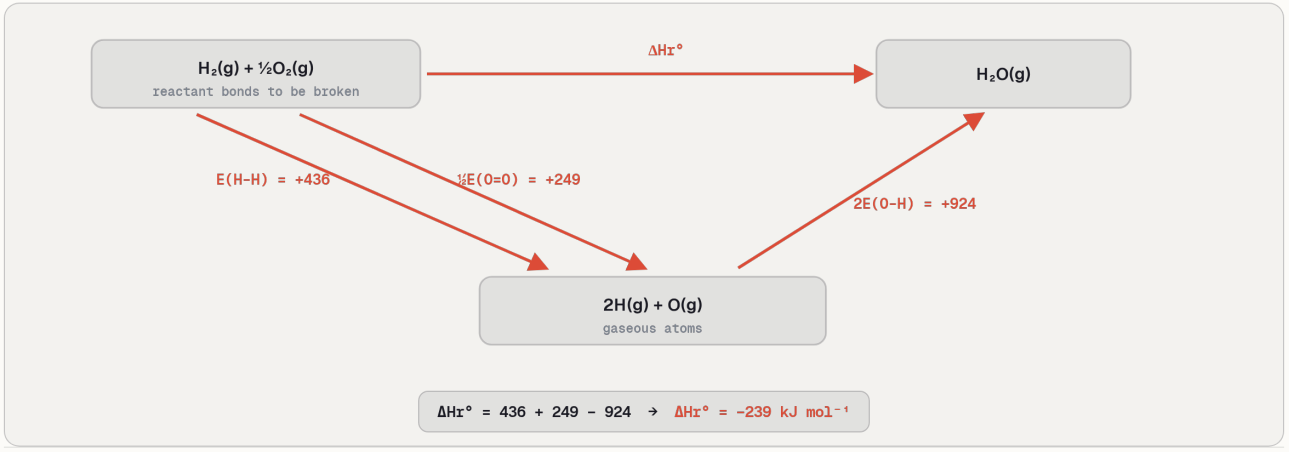
Calculating ΔH from bond enthalpies

$$\Delta H[\text{reaction}] + \Sigma E[\text{products}] = \Sigma E[\text{reactants}]$$

$$\Delta H[\text{reaction}] = \Sigma E[\text{reactants}] - \Sigma E[\text{products}]$$

Bonds broken (reactants) take energy in; bonds formed (products) release energy

SPEC 6.9 - 6.10 · BOND-ENTHALPY CYCLE **Formation of gaseous water through gaseous atoms** route through E(bonds)



FORMULA ΔH[reaction] = Σ E[reactants] – Σ E[products] — break all the reactant bonds, then form all the product bonds. Bond enthalpies are measured in the gaseous state only, and they are mean values — so the result differs from the true measured ΔH.

FIG — *The cycle runs through **gaseous atoms**: break all the reactant bonds, then form all the product bonds.*

WORKED EXAMPLE **Formation of gaseous water**

PROBLEM

Use E(O–H) = +462, E(H–H) = +436 and E(O=O) = +498 kJ mol⁻¹ to find ΔHr for $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$.

1. Apply the cycle: ΔHr + 2E[O–H] = E[H–H] + ½E[O=O].
2. ΔHr = (436) + (½ × 498) − (2 × 462).
3. ΔHr = 436 + 249 − 924.

ANSWER **ΔHr = −239 kJ mol⁻¹**

TIP

Link to atomisation (6.4 v)

For a **diatomic** molecule, bond enthalpy = 2 × ΔatH°: $Cl_2(g) \rightarrow 2Cl(g)$ is +242 kJ mol⁻¹, while $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$ is +121 kJ mol⁻¹.

SPEC 6.10 ≈ 20 MIN

Finding Mean Bond Enthalpies from ΔH

THE SPECIFICATION SAYS

be able to calculate mean bond enthalpies from enthalpy changes of reaction

The bond-enthalpy cycle run in reverse.

The bond-enthalpy cycle runs **both ways**. If the enthalpy change of a reaction is known — from experiment, or from ΔfH°/ΔcH° data — together with **all but one** of the bond enthalpies involved, the unknown **mean bond enthalpy** follows by rearrangement.

$$\Delta H[\text{reaction}] = \Sigma\, E[\text{reactants}] - \Sigma\, E[\text{products}]$$

$$\text{unknown bond broken:}\quad E(\text{unknown}) = \Delta H[\text{reaction}] + \Sigma\, E[\text{product bonds}] - \Sigma\, E[\text{other reactant bonds}]$$

$$\text{unknown bond formed:}\quad E(\text{unknown}) = \Sigma\, E[\text{reactant bonds}] - \Delta H[\text{reaction}] - \Sigma\, E[\text{other product bonds}]$$

The two rearrangements — depending on whether the bond you want is on the reactant (broken) or product (formed) side

- Write the equation, count **every** bond broken and formed (remember molecules like O_2 contain a double bond, counted as one O=O term per molecule).

- Substitute the known values and make the unknown mean bond enthalpy the subject.
- The value obtained is a **mean** for that bond — the same gaseous-state and mean-value limitations (6.9) apply to it.

WATCH OUT

Watch the direction

Bonds **broken** take energy **in**; bonds **formed** release energy. Check your answer against the sign of ΔH: for an exothermic reaction (ΔH negative) the energy released forming product bonds must exceed the energy absorbed breaking reactant bonds.

SPEC 6.11 ≈ 10 MIN

Bond Enthalpy and Reactivity

THE SPECIFICATION SAYS

understand that bond enthalpy data gives some indication about which bond will break first in a reaction, how easy or difficult it is and therefore how rapidly a reaction will take place at room temperature

Which bond breaks first — and how fast the reaction goes at room temperature.

- Bond enthalpy data give an indication of **which bond will break first** in a reaction — the bond with the **lower** bond enthalpy needs **less energy** to break, so it is the easier bond to break.
- They also indicate **how easy or difficult** a reaction is: the larger the bond enthalpy, the more energy is needed to break that bond.
- And therefore they indicate **how rapidly a reaction will take place at room temperature** — reactions whose first step breaks a high-enthalpy bond proceed only slowly without added energy.

EXAM ALERT

The methane values as evidence

The four C–H bonds in CH_4 break with different enthalpies — +423, +480, +425 and +335 kJ mol^{-1}. The spread is exactly why mean values are quoted, and why a bond-enthalpy calculation only approximates the true ΔH of a reaction.

TIP

Connecting to kinetic stability (6.2)

The activation energy seen on a reaction profile is the energy needed to break bonds on the way to reaction — so strong bonds (high bond enthalpies) mean a difficult first step, which is what makes some exothermic reactions kinetically stable.

END OF TOPIC 06

01 · Formulae, Equations and Amount of Substance
02 · Atomic Structure and the Periodic Table
03 · Bonding and Structure
04 · Introductory Organic Chemistry and Alkanes
05 · Alkenes
07 · Intermolecular Forces
08 · Redox Chemistry and Groups 1, 2 and 7
09 · Introduction to Kinetics and Equilibria
10 · Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

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